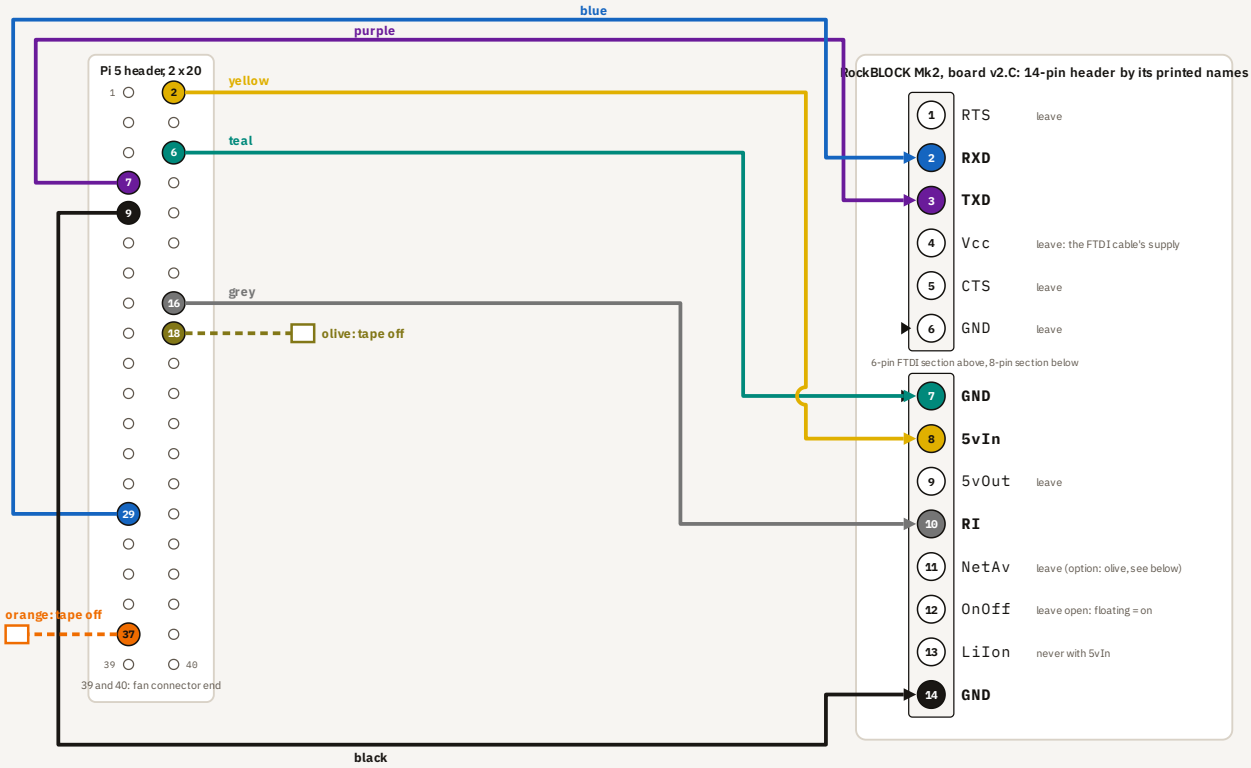
V1 kit: Mk2 on the 9704 harness

23 October 2026 · rev 2
Bench sheet. Prototype, not a production drawing.

Same eight wires, six used, two taped off. The harness stays on the Pi header as it is; its Dupont ends move from the 9704's 16-way header to the Mk2's 14-pin header. Find each Mk2 pin by the name printed beside it. Colour names are written on every wire, so this sheet works in black and white.



Pi header seen from above, pin 1 and 2 at the top end, the fan connector at the 39 and 40 end. Mk2 header in its printed order, RTS at one end and GND at the other; the two printed triangles mark the GND pins at the join of the sections. Arrow: the wire's end at the modem. The small arc is a crossing, not a joint.

WIRE	PI PIN	SIGNAL AT THE PI	MK2 PIN	NOTE
yellow	2	5 V	**8 5vIn**	4.5 to 5.4 V, 450 mA limit. Never together with 13 LiIon. 4 Vcc is the FTDI cable's supply: leave it.
teal	6	GND	**7 GND**	the printed triangle marks it
black	9	GND	**14 GND**	the harness's second ground, used
purple	7	UART2 TX, BCM 4	**3 TXD**	TXD is the modem's input
blue	29	UART2 RX, BCM 5	**2 RXD**	RXD is the modem's output, 3.3 V
grey	16	BCM 23, input	**10 RI**	ring indicator, wired but OFF in software (RI_PIN=0): the line picks up noise, each edge counted as a ring alert; ring alerts come over the UART instead
olive	18	BCM 24, held LOW by the kit	**none: tape it**	LOW on 12 OnOff is sleep; on 11 NetAv it fights the modem's output
orange	37	BCM 26	**none: tape it**	12 OnOff stays open: floating means on

BEFORE POWER

- 5vIn and LiIon never both. On 5vIn the Mk2 charges its supercapacitor for about 20 to 25 s, then the 9602 wakes by itself.
- OnOff left open means on. No Pi pin drives it. Grey may stay on 10 RI; nothing reads it until the Mk2's RI drive is metered.
- Leave open on the Mk2: 1 RTS, 4 Vcc, 5 CTS, 6 GND, 9 5vOut, 11 NetAv, 12 OnOff, 13 LiIon. Flow control is off in the Bridge.
- The 9704 stays unplugged: one modem on UART2 at a time. Serial is 3.3 V both sides, no level shifter.
- **Account:** line rental, a delivery group posting to the Hub, and a Hub device record, before the first session.

THEN IN SOFTWARE, PER KIT

- `/srv/meshsat/.env`: `MESHSAT_IRIDIUM_PORT=/dev/ttyAMA2`, `MESHSAT_IRIDIUM_RI_PIN=0` (rev 2: with 23 the line raised false ring alerts on both kits), NetAv and OnOff pins 0, `MESHSAT_IMT_PORT=auto`.
- `docker compose up -d`: a .env change needs the recreate, not a reboot. Disable `iridium_imt`, enable `iridium`. SBD runs at 19200.
- NetAv option: drop the `24=0` hold from `meshsat-gpio-init.sh`, olive to 11 NetAv, `MESHSAT_IRIDIUM_NETAV_PIN=24`.

Sources: Rock Seven, RockBLOCK v2 Developer Guide 1.4 (14 Sep 2016): supply, sleep pin, signal directions; Ground Control, RockBLOCK 9602/9603 connectors page (the 9602 14-pin table); harness colours from the owner's photos of 21 Sep 2026 (meshsat-fieldkit v1/BUILD.md section 6); the kits' live .env and GPIO service, read 23 Oct 2026. · MeshSat, Leiden · hello@meshsat.net